Practice workbook · Surds: simplifying and rationalising the denominator

Exact surd arithmetic without a calculator: taking square factors out of roots and clearing surds from denominators.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-LOG-04

1. Question 1 · part whole · 4 marks

2023 NOV · Paper 01

Worked-example retry · Attempt part whole only.

1

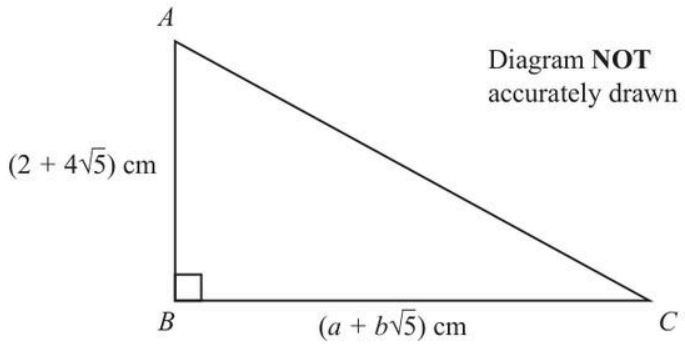


Figure 1

Figure 1 shows the triangle ABC

$\angle ABC = 90°$ $AB = \left(2+4\sqrt{5}\right)$ cm $BC = \left(a+b\sqrt{5}\right)$ cm where a and b are integers.

The area of triangle $ABC = \left(34+11\sqrt{5}\right)$ cm^2

Without using a calculator, find the value of a and the value of b

(4)

OLevels.COM — SCHOOL —

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 7 3 5 8 4 A 0 4 3 6

1. Question 1 · part whole · 4 marks

2023 NOV · Paper 01

Worked-example retry · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 1 continued

OLevels.COM — SCHOOL —

(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 9 · part c · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part c only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

2. Question 9 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part whole · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

3 Given that $\frac{a+b\sqrt{5}}{6-2\sqrt{5}} = \frac{9+4\sqrt{5}}{c}$ where a, b and c are prime numbers,

find the value of a, the value of b and the value of c

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 3 · part whole · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

Question 3 continued

OLevels.com SCHOOL

(Total for Question 3 is 5 marks)

3. Question 3 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 1 · part whole · 4 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN

1 Given that $\frac{a+\sqrt{3}}{2-\sqrt{3}} = 11 + b\sqrt{3}$ where a and b are integers,

find the value of a and the value of b.

(4)

OLevels.com — SCHOOL —

(Total for Question 1 is 4 marks)

4. Question 1 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 6 · part a · 2 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 Given that $\dfrac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \dfrac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\dfrac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$

where p, q, r and s are integers.

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 7 1 6 4 1 A 0 1 2 3 6

5. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 1 · part whole · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN

1 Given that $\frac{2\sqrt{3}-4}{3\sqrt{3}+5}$ can be written in the form $a+b\sqrt{3}$ where a and b are integers,

find, without using a calculator, the value of a and the value of b

Show your working clearly.

(3)

OLevels.com — SCHOOL —

(Total for Question 1 is 3 marks)

6. Question 1 · part whole · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 1 · part whole · 4 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

1 Given that $\frac{a-\sqrt{48}}{\sqrt{3}+1}$ can be written in the form $b\sqrt{3}-9$ where a and b are integers,

find the value of a and the value of b
Show your working clearly.

(4)

OLevels.com SCHOOL

(Total for Question 1 is 4 marks)

7. Question 1 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 1 · part whole · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

1 $\left(8-\sqrt{6}\right)w=50$

Without using a calculator, find the value of w

Give your answer in the form $\frac{a+b\sqrt{6}}{c}$ where a and b are integers and c is prime.

Show your working clearly.

(3)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN T

OLevels.com — SCHOOL —

(Total for Question 1 is 3 marks)

8. Question 1 · part whole · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 1 · part whole · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part whole only.

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

1

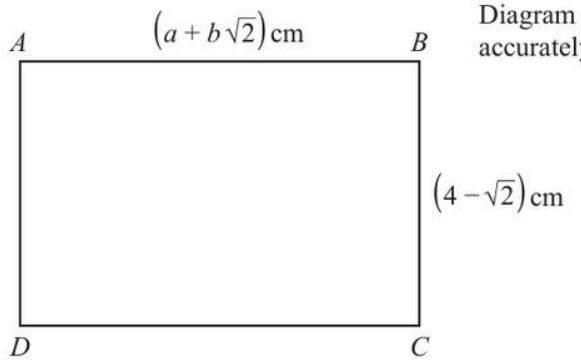


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the rectangle $ABCD$.

$AD = BC = (4 - \sqrt{2})$ cm and $AB = DC = (a + b\sqrt{2})$ cm where a and b are integers.

The area of the rectangle is $(10 + \sqrt{2})$ cm^2

Find the value of a and the value of b
Show your working clearly.

(4)

(Total for Question 1 is 4 marks)

9. Question 1 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 9 · part b · 2 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 9 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 1 · part whole · 5 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

1 Given that $\frac{a+2\sqrt{5}}{3-\sqrt{5}} = \frac{11+b\sqrt{5}}{2}$ where a is an integer and b is prime,

find the value of a and the value of b
Show your working clearly.

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN T

OLevels.com — SCHOOL —

(Total for Question 1 is 5 marks)

11. Question 1 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL